the problem that C-3 goes out of safe voltage envelope and some other cell is left uncharged to its full voltage. Since the battery pack has reached its full voltage, the BMS would have to open its charging FET and stop any charging current.

On the extreme end, at the time of full discharge, as shown in Fig. 4(d), the weak cell (C-3) discharges much faster than other cells, because of which the overall battery voltage reaches its discharge limit before the last cell (C-4) can be discharged fully. This means the battery has spare capacity that cannot be used any more.

In practice, Fig. 4(a) and Fig. 4(b) represent cases which cannot be achieved without active intervention from a BMS, while Fig. 4(c) and Fig. 4(d) represent cases that are almost always found in battery packs.

So, the BMS tries to either expend excess energy from weak cells or redistribute energy within the battery pack to avoid cells going out of their specifications. This process of correcting the distribution of energy is quite literally called balancing. Balancing the cells by expending excess energy is best done either during or just after the charging process, so that cells being expended are also replenished and, if required, the flow of current continues, and thus it is called top-balancing.

Redistribution of energy can be done while charging or while discharging, or while the battery is not being used for any load. So, redistribution allows either top-balancing or bottom-balancing operations. Balancing by expending excess energy is carried out by connecting the weak cells across a burden resistance, which converts the excess energy into heat. The rate of balancing is limited by the resistance value. This is also called passive balancing, because only a connection to the burden resistance is required for top-balancing to happen.

Generally, since the BMS does not know which cells could be the weak

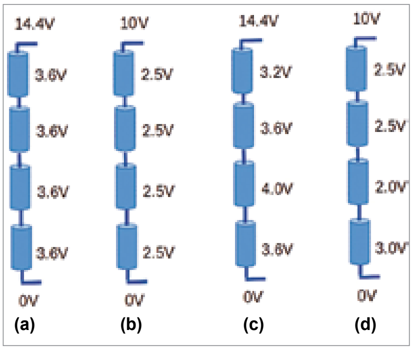

Fig. 4: Various stages of a LiFePO4 battery pack: (a) balanced and charged, (b) balanced and discharged, (c) unbalanced and charged, and (d) unbalanced and discharged

ones in practice, they are designed to provide a burden resistance across each cell in the series connection. The value of the burden resistance is chosen such that balancing happens within the shortest time possible while keeping the resistance temperature within safe limits.

Redistribution of energy is much more complex because, to redistribute energy some form of energy storage device is required, which can swap energy from one cell to the other cell. This is carried out by creating a matrix of capacitors or inductors (transformers) that are then switched across cells to allow for rebalancing. Some innovative designs also use a separate cell itself as the energy swapping container.

Since we use reactive energy storage here and the BMS also needs to actively switch the energy storage devices so that balancing can happen, such an arrangement is called active balancing. It has the advantage that it can be performed at any time during the charge-discharge cycle, and it tries to conserve energy rather than simply expending it has heat.

Many complex schemes are possible when it comes to active balancing. These schemes are out of the scope of this article, so the designers may go through the available research papers to figure out a scheme that is feasible for the design at hand. It suffices to say here that active balancing has the potential to increase the available battery life. But a BMS with active balancing will almost always increase the cost of the battery pack.

Reporting battery condition

Reporting the present state of the battery is an important function of the BMS. The simplest BMSes provide visual feedback on the state of charge of the battery by using LED based indicators (LED bars or individual LEDs). Some more advanced BMSes may have a screen or a touch-screen interface. This allows the user to not only monitor the live parameters of the BMS but also easily setup the BMS parameters at the time of connection to the battery. This suffices in most stationary applications.

A BMS that goes into the industrial setting may use the RS232 or RS485 physical bus and a simple protocol, such as MODBUS, to allow an external supervisory control unit to read and write parameters. Some others may allow this information exchange to happen over Bluetooth (so that your smartphone can watch the battery over an app) or make it available over IP networks, either via regular Ethernet ports or a Wi-Fi port.

If the battery is being used in an electric vehicle of some kind, the vehicle's main electronic control unit (ECU) will expect the BMS to make all the relevant information available in some kind of known digital format. The most common data connection scheme used in the automotive world is the CAN bus. Accordingly, the BMS may be required to make its data available over the CAN bus. Setting up the BMS with all its parameters may also be done over the same connection.

The reporting arrangement used by the BMS directly affects the cost of the BMS, with more exotic schemes making the BMS more expensive. As always, the designer must exercise choice and trade-off to choose the feature that is most suited for the project at hand. EFY